



Problem

Given an integer array of size n , find the majority element. A majority element is an element that appears more than $\lfloor n/2 \rfloor$ times.

If a majority element exists, print it. Otherwise, print -1.

Complete the function `majorityElement` that takes the following parameter:

`int arr[n]`: an array of integers Returns `int`: the majority element if it exists; otherwise -1.

Submissions

Input Format

The first line contains an integer n , the number of elements in the array. The second line contains n space-separated integers.

Constraints

$1 \leq n \leq 10^5$ $-10^9 \leq arr[i] \leq 10^9$

Output Format

Print the majority element if it exists; otherwise, print -1.

Leaderboard

Discussions

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Language

Python 3



```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 majority_element = 0
4 count = 0
5 for num in arr:
6     if count == 0:
7         majority_element = num
8         count = 1
9     elif num == majority_element:
10        count += 1
11    else:
12        count -= 1
13 count = 0
14 for num in arr:
15     if num == majority_element:
16         count += 1
17 if count > n // 2:
18     print(majority_element)
19 else:
20     print(-1)
```

Line: 2 Col: 38

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Test case 0

Compiler Message

Test case 1

Success

Hidden Test Case

Use print or log statements to debug why your hidden test cases are failing. Hidden test cases are used to evaluate if your code can handle different scenarios, including corner cases.